

Chenhao Sun

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EDUCATION

University of British Columbia | B.A.Sc., Engineering Physics (Expected 2029)

EXPERIENCE

UBC Orbit, Mechanical Design Developer | Sept. 2025 – Present

Student design team building a flight-qualified 1U CubeSat that lets radio amateurs request and downlink imagery of chosen locations, with applications to disaster relief; launch planned 2028

- Led structural design iterations for the antenna deployer module, developing and refining mechanical solutions in Onshape
- Designed a burn-wire door release in which a Dyneema line holds the door shut and restrains the coiled antennas; added mechanical stops to limit spring deflection and keep wire tension consistent
- Designed a metal bracket to restrain a balun within a 2 mm height budget
- Running plastic deformation tests on antenna material across 6 coil radii to select the tightest radius that still meets maximum deflection requirements
- Supported vibration and thermal vacuum testing of the module at ESA facilities; found issues with antenna deployment in TVAC due to increased friction, leading to redesign with antenna release mechanism and shape of constraining walls
- Worked with the electrical and controls teams to fit wiring and PCB layouts around mechanical constraints
- Assembled flight hardware in cleanroom conditions following ESA cleanliness guidelines

PROJECTS

Mars Habitat Robot | July 2026 – Aug. 2026

Autonomous robot built for a robotics competition; completed 3 course tasks (metal-rock retrieval, teletubby detection, solar panel uncovering)

- Designed, modelled, and manufactured a 3-DOF arm (laser-cut Delrin), with a 3D printed claw, and shaft collars, sizing links and selecting servos from torque and reach calculations
- Redesignated pinned shaft couplings as screw-fastened joints after testing showed excessive deflection, eliminating visible deflection
- Helped build a Colpitts-oscillator metal detector and integrated a K230 vision module for rock, teletubby, and solar panel detection
- Helped design the stepper motor driver PCB in Altium and soldered the team's other boards
- Tuned PID for object alignment; helped trace control instability to motor EMI and fixed it with a protoboard hardware-debounce circuit

Metal Cellphone Stand | Jan. 2026 – Apr. 2026

- Designed and machined an angle-adjustable cellphone stand from aluminum and brass using a lathe, mill, drill press, and tapping tools
- Held ± 0.5 mm tolerances on mating parts for smooth, repeatable adjustment
- Learned facing, turning, threading, edge finding, and face and side milling, and worked from detailed drawings

TECHNICAL SKILLS

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- Mechanical: CAD (Onshape, Solidworks, AutoCAD); Machine shop operation
 - Electrical: Altium, Circuit Design & Analysis; Oscilloscopes, Multimeters, Arduino
 - Software: C; C++; Java; Python; Git; Jupyter Notebook; LaTeX; MatLab; Excel; Google Sheets

